

Formation, structure and antibacterial activities of silazane networks grafted with poly(ethylene glycol) branches



François-Xavier Perrin^{a,*}, Thi Dieu Hang Nguyen^b, Dinh Lam Nguyen^b

^a Laboratoire MAPIEM EA 4323, SeaTech-Ecole d'ingénieurs, Université de Toulon BP 20132, 83957 La Garde Cedex, France

^b Danang University of Science and Technology, University of Danang, 54 Nguyen Luong Bang, Danang, Viet Nam

ARTICLE INFO

Article history:

Received 29 October 2014

Received in revised form 12 June 2015

Accepted 22 June 2015

Available online 17 July 2015

Keywords:

Silazane

Poly(ethylene glycol)

Anti-bacterial

ABSTRACT

The aim of this investigation was to develop coating materials based on poly(ethylene glycol) (PEG) covalently grafted onto silazane polymers for marine antifouling applications. The optimum conditions for grafting PEG were defined to have a high selectivity toward olefin hydrosilylation. Thick crack-free films were obtained by curing at room temperature of the PEG grafted silazane precursors. The solidification process has been investigated by FTIR spectroscopy, ²⁹Si NMR in the solid state, thermogravimetric analysis (TGA) as well as elemental analysis. The main reactions that occur during curing are hydrolysis-condensation reactions of alkoxy silane, Si–H and Si–N functionalities. The PEG-graft-PSZ coatings exhibit excellent repellency against gram-negative *Neisseria* sp. and gram-positive *Clostridium* sp. in comparison with the pristine polysilazane surface. The anti-adhesion performance of the coatings depends on the grafting density and the chain length of PEG. The shortest PEG(350 g/mol)-graft-PSZ with the highest graft density was found to have the best anti-adhesion performance. As the density of grafted PEG(750 g/mol) and PEG(2000 g/mol) chains onto the PSZ surface is approximately equal, the relative effectiveness of these two types of PEG is controlled by the length of the PEG chain. The PEG(2000 g/mol)-graft-PSZ coatings are more efficient than the PEG(750 g/mol)-graft-PSZ coatings for the bacterial anti-adhesion.

© 2015 Elsevier B.V. All rights reserved.

1. Introduction

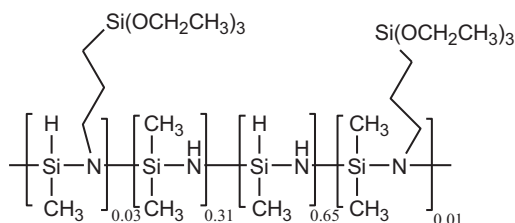
Bacterial adhesion and marine biofouling on a surface may lead to increased operational and maintenance costs [1] or can be critical to the performance of a biomedical device [2]. Thus, a great attention has been paid to the development of protein resistant or non-fouling surfaces for a variety of biomedical devices [2] as well as for marine applications [3]. A coating of polyethylene glycol (PEG) is currently the most common approach to inhibit bioadhesion. The PEG grafted surface become difficult to be approached by incoming particles due to the high mobility, large exclusion volume and steric hindrance effect of the highly hydrophilic layer. For examples, glass [4], silicon [5,6], gold [7,8] or polymer [8–14] surfaces modified with PEG brushes and UV crosslinked PEG diacrylate solid membranes [6,15] are able to resist the adhesion of many proteins [5,9–11,16,17] and bacteria [4,6,8,14,18–21]. Furthermore, excellent fouling release properties relative to marine algal fouling were reported for surface active block copolymers with amphiphilic side chains [22,23]. Amphiphilicity was imparted

by a hydrophobic aliphatic [22] or fluorinated [23] group combined with a hydrophilic PEG group. As a novel approach to functionalize oxide surfaces, PEG grafted on a titanium surface was used for its antiadhesive properties and for the covalent immobilization of an antimicrobial peptide [20]. Polysilazane polymers consist of backbones with alternating Si–N bonds and are widely used as precursors of silicon carbonitride ceramics [24–27]. The polymer-to-ceramic transformation is generally performed at high temperatures (above 1000 °C). On the other hand, alkoxy silyl substituted polysilazanes can be crosslinked by the reaction with atmospheric moisture to form a cured surface film on the substrate surface [28,29]. Motivated by cost, environmental concerns and regulations to limit the volatile organic compound (VOC) content, the paint and coating industries are increasingly shifting toward aqueous based systems, powder coatings or UV curable coatings. Moisture cured polysilazane coatings provide an interesting alternative to other standard technologies when they need low VOC and low energy consumption.

The purpose of this work was to construct new antifouling coatings by covalent grafting of methoxy-terminated PEG chains onto polysilazane polymers through a highly stable silicon–carbon bond. Prime et al. [30] reported that replacing the terminal –OH of oligo(ethylene oxide) by –OCH₃ did not decrease the ability

* Corresponding author.

E-mail address: perrin@univ-tln.fr (F.-X. Perrin).



Scheme 1. Molecular structure of PSZ.

of the surface to resist protein adsorption. Recently, preliminary results concerning the grafting of PEG chains onto oligosilazane were reported [31]. In this paper, a careful investigation of the network building from the PEG grafted silazane precursors is reported. Besides, the morphology of PEG chains in the cured material and the thermal stability of the surfaces are investigated. We have examined the effects of the length and grafting density of PEG chains upon bacteria adhesion. In addition to antibacterial activity, these coatings showed strong adherence to metal substrates (steel and aluminum) and the high crosslinked density of the moisture cured polymers confer high mechanical strength and good barrier properties. Altogether, these properties offer promising potential in terms of a one-coat system combining anticorrosive and antibacterial properties.

2. Experimental

2.1. Materials

Monomethoxy poly(ethylene glycol) (MPEG) with the average molecular weights of 350 g/mol, 750 g/mol and 2000 g/mol and Karstedt's catalyst were purchased from Sigma–Aldrich. All three types of MPEG were dried under vacuum before usage. The polysilazane precursor, polydimethylhydrosilazane containing triethoxysilanes was provided by the Clariant Company. The molecular structure of the polysilazane precursor is described on Scheme 1.

PSZ was used as received without any further purification. Allyl bromide (Allyl-Br) purchased from Acros was distilled under a nitrogen atmosphere in the dark and subsequently kept in darkness before usage. Allyl-Br is sensitive to light and may polymerize on exposure to light.

All synthetic procedures were performed under a dry argon atmosphere unless otherwise stated.

2.2. Synthesis

Allyl-terminated monomethyl polyethylene glycol (Allyl-PEG) were synthesized by the reaction of MPEG with an excess amount of allyl bromide. As a representative example, allyl bromide (60.5 g, 497 mmol), MPEG350 (25 g, 71 mmol) and sodium hydroxide (6 g, 150 mmol) were heated at 70 °C for 24 h. The purification protocol included the following steps: separation of sodium hydroxide by filtration, recovery of allyl-PEG trapped in sodium hydroxide by washing with chloroform, removal of allyl bromide and chloroform in a rotary evaporator, removal of unreacted MPEG traces by extraction in water/chloroform, removal of chloroform in a rotary evaporator and, finally vacuum drying of allyl-PEG for 12 h at 50 °C.

The yield of synthesis of allyl-PEGs is about 95%. Allyl-PEGs obtained were characterized by ^1H NMR and ^{13}C NMR to confirm their structure. **Allyl-PEG350:** ^1H NMR (CDCl_3): δ (ppm) 3.34 [s, 3H, $\text{CH}_3\text{-O}$], 3.4–3.8 [m, 29H, $-\text{CH}_2\text{-CH}_2\text{-O}$], 3.99 [d, 2H, $\text{CH}_2=\text{CH}-\text{CH}_2\text{-O}$], 5.22 [m, 2H, $\text{CH}_2=\text{CH}-$], 5.87 [m, 1H, $\text{CH}_2=\text{CH}-$]. ^{13}C NMR (CDCl_3): δ (ppm) 58.41 [$\text{CH}_3\text{-O}$], 68.64 [$-\text{CH}_2\text{-CH}_2\text{-O}-\text{CH}_3$], 69.78 [$-\text{CH}_2\text{-CH}_2\text{-O}-$],

71.13 [$-\text{CH}_2\text{-CH}_2\text{-O}-\text{CH}_3$], 71.36 [$\text{CH}_2=\text{CH}-\text{CH}_2-$], 116.18 [$\text{CH}_2=\text{CH}-\text{CH}_2-$], 134.02 [$\text{CH}_2=\text{CH}-\text{CH}_2-$]. **Allyl-PEG750:** ^1H NMR (CDCl_3): δ (ppm) 3.36 [s, 3H, $\text{CH}_3\text{-O}$], 3.4–3.8 [m, 67H, $-\text{CH}_2\text{-CH}_2\text{-O}$], 4.01 [d, 2H, $\text{CH}_2=\text{CH}-\text{CH}_2\text{-O}$], 5.23 [m, 2H, $\text{CH}_2=\text{CH}-$], 5.89 [m, 1H, $\text{CH}_2=\text{CH}-$]. ^{13}C NMR (CDCl_3): δ (ppm) 57.94 [$\text{CH}_3\text{-O}$], 68.53 [$-\text{CH}_2\text{-CH}_2\text{-O}-\text{CH}_3$], 69.63 [$-\text{CH}_2\text{-CH}_2\text{-O}-$], 70.99 [$-\text{CH}_2\text{-CH}_2\text{-O}-\text{CH}_3$], 71.10 [$\text{CH}_2=\text{CH}-\text{CH}_2-$], 115.78 [$\text{CH}_2=\text{CH}-\text{CH}_2-$], 133.99 [$\text{CH}_2=\text{CH}-\text{CH}_2-$]. **Allyl-PEG2000:** ^1H NMR (CDCl_3): δ (ppm) 3.37 [s, 3H, $\text{CH}_3\text{-O}$], 3.4–3.8 [m, 194H, $-\text{CH}_2\text{-CH}_2\text{-O}$], 4.01 [d, 2H, $\text{CH}_2=\text{CH}-\text{CH}_2\text{-O}$], 5.24 [m, 2H, $\text{CH}_2=\text{CH}-$], 5.90 [m, 1H, $\text{CH}_2=\text{CH}-$]. ^{13}C NMR (CDCl_3): δ (ppm) 58.55 [$\text{CH}_3\text{-O}$], 68.98 [$-\text{CH}_2\text{-CH}_2\text{-O}-\text{CH}_3$], 70.12 [$-\text{CH}_2\text{-CH}_2\text{-O}-$], 71.48 [$-\text{CH}_2\text{-CH}_2\text{-O}-\text{CH}_3$], 71.72 [$\text{CH}_2=\text{CH}-\text{CH}_2-$], 116.54 [$\text{CH}_2=\text{CH}-\text{CH}_2-$], 134.36 [$\text{CH}_2=\text{CH}-\text{CH}_2-$].

Grafting of allyl-PEG molecules onto the polysilazane (PSZ) chain was performed by using a hydrosilylation reaction between the Si–H group of PSZ and the terminal C=C bond of the allyl-PEG with an excess amount of PSZ in presence of Karstedt's catalyst. The catalyst Pt (dvs) was added into a three-necked flask containing an amount of allyl-PEG ([Pt]/[allyl-PEG] molar ratios of 1×10^{-3} or 3×10^{-3}). After stirring for 15 min, PSZ was added ([Si–H]/[allyl-PEG] molar ratios of 7, 10, 16.5 or 26.5). The reaction occurred under argon or dry air (in this case, the reaction flask was equipped with a long glass fitting containing anhydrous calcium chloride). The reaction proceeded with stirring at a temperature of 70, 80 or 85 °C. The taking of samples was regularly carried out for tracking the kinetics of reaction. We define that a, b, c are the different protons and carbon atoms on the allyl branch of the PEG molecule, whereas a', b', c' are the different protons and carbon atoms of the same branch after being grafted onto the PSZ chain. Different Si–H/allyl molar ratios of 10, 16.5 and 26.5 were investigated for all three types of allyl-PEG. The grafted products are symbolized as PSZ–PEGX–Y with X being the molecular weight of MPEG and Y being the Si–H/allyl molar ratio. The products obtained were colorless or slightly yellow viscous oils (from PEG350), pastes (from PEG750) or solids (from PEG2000) which are soluble in aromatic solvents.

Thin films (ca. 1 μm thick) were prepared by dip-coating technique from 10% ethyl acetate (for PSZ and PSZ–PEG350) or chloroform (for PSZ–PEG750 and PSZ–PEG2000) solutions using a substrate dipping speed of 240 mm/min and withdrawal speed of 120 mm/min. The substrates used were degreased aluminum plates (2.5 cm $\times$ 5 cm $\times$ 0.1 cm). The films used to track the crosslinking reaction course were dried at 21–22 °C and 50% humidity. Free and supported 60 μm thick films were obtained from PSZ, PSZ–PEG350 or a chloroform:toluene (50:50, v:v) solution of PSZ–PEG750 and PSZ–PEG2000 using an automatic film applicator Sheen 1137. Thick films were cured for 7 days at 25 °C and 84% humidity, which is the optimum cure time as determined from hardness measurements according to the Persoz method (Fig. S1 in supporting information). The methanol extraction of the isomers of allyl-PEG and/or unreacted allyl-PEG was followed by FTIR spectrometry in transmission mode on free films. It took about 10 min, 20 min and 3 h to extract ungrafted PEG from PSZ–PEG350, PSZ–PEG750 and PSZ–PEG2000, respectively. As-prepared PEG grafted PSZ films are translucent while they are transparent in the naked eye after extraction of free PEG chains (Fig. S2 in supporting information).

The conversions of Si–H, NH and C=C were calculated using the following equations:

$$\%C_{\text{Si-H}} = \frac{\frac{A_{\text{Si-H(o)}}}{A_{\text{Si-CH}_3(\text{o})}} - \frac{A_{\text{Si-H(t)}}}{A_{\text{Si-CH}_3(\text{t})}}}{\frac{A_{\text{Si-H(o)}}}{A_{\text{Si-CH}_3(\text{o})}}} \times 100$$

$$\%C_{N-H} = \frac{\frac{A_{N-H(o)} - A_{N-H(t)}}{A_{Si-CH_3(o)}} - \frac{A_{N-H(o)}}{A_{Si-CH_3(o)}}}{\frac{A_{N-H(o)}}{A_{Si-CH_3(o)}}} \times 100$$

$$\%C_{C=C} = \frac{\frac{A_{C=C(o)} - A_{C=C(t)}}{A_{Si-CH_3(o)}} - \frac{A_{C=C(o)}}{A_{Si-CH_3(o)}}}{\frac{A_{C=C(o)}}{A_{Si-CH_3(o)}}} \times 100$$

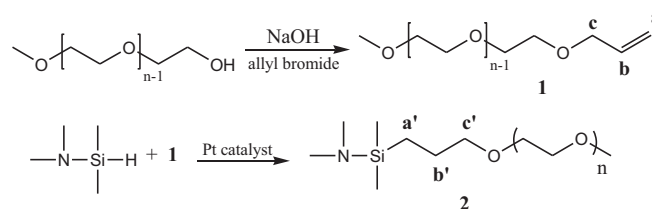
where $A_{Si-CH_3(o)}$, $A_{Si-CH_3(t)}$ are the absorption bands of Si–CH₃ at 1253 cm^{−1} initially and at time t , respectively, $A_{Si-H(o)}$ and $A_{Si-H(t)}$, the absorption band of Si–H at 2125 cm^{−1} initially and at time t , respectively, $A_{N-H(o)}$ and $A_{N-H(t)}$, the absorption bands of N–H at 3384 cm^{−1} initially and at time t , respectively and $A_{C=C(o)}$, $A_{C=C(t)}$, the absorption bands of C=C at 1592 cm^{−1} initially and at time t , respectively.

2.3. Characterization methods

Average molecular weights were determined by size exclusion chromatography using three μ styragel columns (HR0.5, HR1 and HR2) calibrated with polystyrene standards and with THF as eluent (1 mL/min flow rate). The ¹H spectra were recorded on a Bruker Avance 400 (¹H at 400 MHz), at 298 K. The ²⁹Si NMR spectra were recorded on a Bruker DSX400 spectrometer operating at 79.501 MHz for ²⁹Si nuclei. Magic angle spinning (MAS) NMR spectra were acquired at a rotation frequency of 10 kHz. The repetition delay and pulse width were 40 s and 4 μ s, respectively. The repetition delay, contact time and pulse width were 5 s, 5 ms and 3.5 μ s (¹H 90° pulse), respectively, for ²⁹Si CP MAS NMR. In ²⁹Si and ¹H NMR, the compound used as chemical shift reference was tetramethylsilane. ²⁹Si NMR spectra were deconvoluted by the Origin software by considering a Gaussian shape of individual peaks (Fig. S4 in supporting information). The terminology used for the description of the different types of Si atoms incorporates the use of the letters *M*, *D* and *T* which represent R₃Si(OSi)_{1−*x*}(OR')_{*x*}, R₂Si(OSi)_{2−*x*}(OR')_{*x*} and RSi(OSi)_{3−*x*}(OR')_{*x*} units, respectively with R' = Et or H. In addition, superscripts refer to substituents on silicon other than Me, and underscript refer to the number of –OSi groups bound to the silicon atom of interest. For example, D₂^N indicates the unit MeNSi(OSi)₂. DSC measurements were carried out on a DSC Q100 differential scanning calorimeter in nitrogen atmosphere at heating/cooling rates of 10 K min^{−1}. The glass transition temperature (*T*_g) and melting properties were determined from the second heating scan. Thermogravimetric analysis (TGA) was carried out using a TA instruments SDT Q600 under a flowing atmosphere of argon or air (both at 100 mL/min) using a heating rate of 10 K min^{−1}. The elemental analyses were carried out in the analysis laboratory of the elemental analysis department at CNRS Vernaison, France.

2.4. Protocol of the marine bacterial adhesion

Three marine bacteria, bacillus *Clostridium* sp. SR1, micrococcus *Neisseria* sp. LC1 and micrococcus *Neisseria* sp. SC1, were used for the adhesion studies. Two types of culture media were tested: Vätäänen nine-salt solution (VNSS) and Sapec-peptone solution. The PSZ and PSZ-PEG coatings were previously sterilized in methanol, immersed in culture media that contained the bacteria and finally incubated at 30 °C for 24 h. Then coatings were rinsed in sterilized phosphate buffered saline to remove the non-adherent bacteria. The adhesive bacteria were removed from the surfaces using an ultrasonic cleaner for 5 min and re-incubated at 30 °C for 24 h on an agar plate that contained culture media. The



Scheme 2. Synthesis of polysilazane based amphiphile.

number of adhered bacteria (CFU/cm²) was determined using a Colony Counter BZG30.

3. Results and discussion

3.1. Hydrosilylation of allyl PEG with polysilazane

Synthesis of the novel polysilazane-based amphiphile **2** was achieved via two steps as shown in Scheme 2. Allyl methoxy PEG **1** was synthesized through the reaction of the hydroxyl end group of monomethyl end-capped PEG with an excess amount of allyl bromide [32]. The hydrophilic PEG segment was coupled to polysilazane by the Karstedt's catalyzed hydrosilylation reaction as Karstedt's catalyst was found to be efficient at low concentration (5–25 ppm) in hydrosilylation of allyl polyethers by siloxane polymers with active SiH bonds [33].

The progress of the hydrosilylation reaction was followed by ¹H NMR spectroscopy (Fig. 1). The structure of grafted group=Si–CH₂–CH₂–CH₂–O– of PSZ-PEG is very close to that of the group=Si–CH₂–CH₂–CH₂–N– of the PSZ precursor. The resonance peaks of the protons **H_{a'}** and **H_{b'}** of PSZ-PEG therefore appear at chemical shifts very similar to that of the protons =Si–CH₂–CH₂– (0.62 ppm) and =Si–CH₂–CH₂– (1.5 ppm) of PSZ, respectively. Likewise, the chemical shift of the proton **H_{c'}** of PSZ-PEG is very close to that of protons –O–CH₃ and –O–CH₂–CH₂–O– of PEG, situated at around 3.4 ppm. For overcoming these complications, the kinetic investigation of the hydrosilylation reaction was accomplished by following the decrease of the allyl-proton resonances. The absence of a specific signal at 0.98–1.12 ppm (SiCH₂CH₃) suggests that the Markovnikov α -addition product is formed, if any, in a limited extent. In the first few hours, the decrease of the allyl protons at δ = 5.89 ppm, δ = 5.24 ppm and δ = 5.15 ppm is concomitant with the decrease of the SiH protons at δ = 4.64 ppm and δ = 4.39 ppm, which confirms

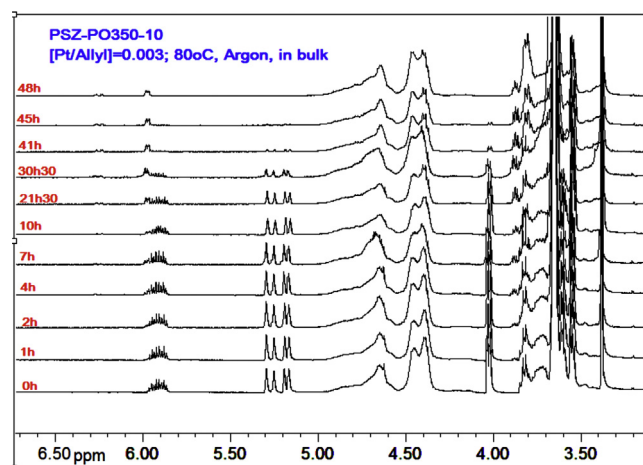
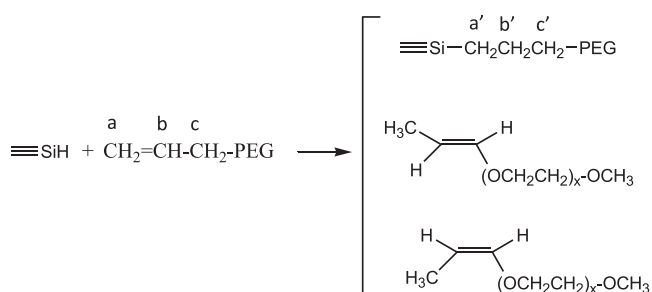


Fig. 1. ¹H NMR spectra taken over the course of the allyl-PEG350 and PSZ reaction in solvent free and argon atmosphere conditions (Si–H/allyl = 10, *T* = 80 °C).

Table 1

Effect of grafting conditions on the conversion of allyl-PEG350 (SiH/allyl = 10).

Grafting conditions				Sign of reaction	Time of reaction (h)	Conversion of allyl (%)		
Atmosphere	Temp (°C)	Pt/allyl (mol)	Solvent			C _{AP,total}	C _{AP,hydro}	C _{AP,iso}
Argon	80	0.003	Solventless	R ₁	48	100	71.8	28.2
Dry air	80	0.003	Solventless	R ₂	71	100	65.7	34.3
Argon	80	0.001	Solventless	R ₃	78	78.6	58.7	19.9
Dry air	80	0.001	Solventless	R ₄	120	94.1	67.7	26.4
Argon	85	0.001	Solventless	R ₅	78	88	58.8	29.2
Argon	70	0.003	Solventless	R ₆	78	88.3	70.7	17.6
Argon	80	0.003	Toluene	R ₇	78	74.1	51.1	23.0

**Scheme 3.** Products formed during hydrosilylation of allyl-PEG with PSZ.

the formation of the hydrosilylation product. Afterwards, the intensity of allyl protons still decrease but two new proton signals at $\delta = 5.95$ ppm and $\delta = 6.2$ ppm concomitantly appear and increase in intensity. These last signals are attributed to the internal isomers cis and trans, respectively, of the propenyl ether group, which result

from the isomerization of allyl-PEG [34]. The allyl-PEG isomerization was therefore considered as a secondary reaction that takes place simultaneously to hydrosilylation (Scheme 3).

The optimum conditions for grafting of PEG were studied, changing the temperature of reaction medium, type of atmosphere and reaction medium and amount of catalyst (Table 1). The total conversion of Allyl-PEO (%C_{AP,total}), the conversion of allyl-PEO associated with isomerization (%C_{AP,iso}) and the conversion of allyl-PEO associated with hydrosilylation (%C_{AP,hydro}) were calculated according to the following formulas:

$$\%C_{AP,total} = \frac{I_{CH_2^-(o)} - I_{CH_2^-(t)}}{I_{CH_2^-(o)}} \times 100$$

$$\%C_{AP,iso} = \frac{2 \times (I_{CH,cis} + I_{CH,trans})_{(t)}}{I_{CH_2^-(o)}} \times 100$$

$$\%C_{AP,hydro} = \%C_{AP,total} - \%C_{AP,iso}$$

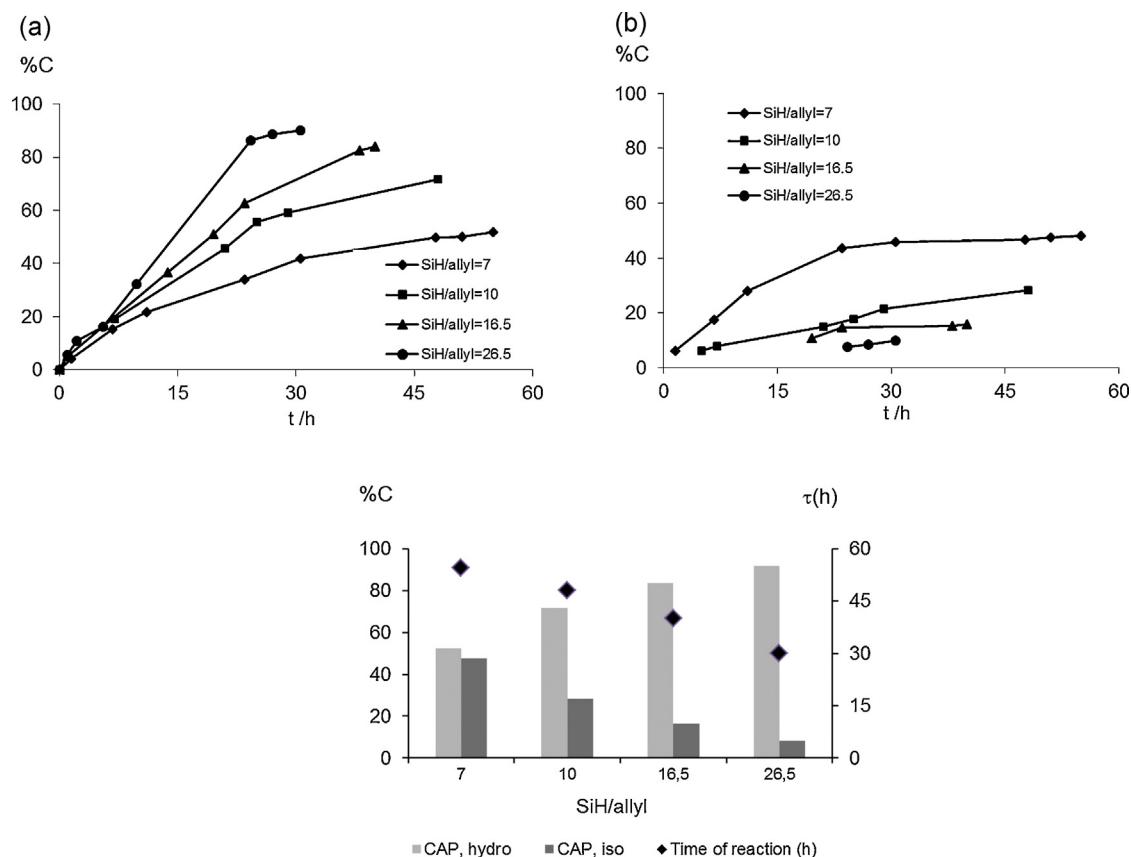
**Fig. 2.** Conversion of allyl-PEG350 for hydrosilylation (C_{AP,hydro}) (a,c) and isomerization (C_{AP,iso}) (b,c) as a function of reaction time (a,b) or at reaction end (c) and reaction time to achieve 100% conversion in allyl functionalities (c).

Table 2

Conversion of allyl-PEG and density of the grafted PEG onto PSZ chains calculated from ^1H NMR spectra.

$X = \text{M}_{\text{MPEG}}$ (g/mol)	$Y = \text{SiH/allyl}$	t (h)	$\%C_{\text{AP,total}}$	$\%C_{\text{AP,hydro}}$	$\%C_{\text{AP,iso}}$	δ^a
350	7	54.5	100	52.5	47.5	0.75
	10	48	100	71.8	28.2	0.72
	16.5	40	100	83.6	16.4	0.51
	26.5	30	100	91.8	8.2	0.35
750	10	55	96.7	62.8	33.9	0.63
	16.5	55	91.6	68.0	23.6	0.41
	26.5	55	97.3	88.9	8.5	0.34
2000	10	55	80.1	59.9	20.2	0.60
	16.5	55	72.6	61.9	10.7	0.38
	26.5	55	86.6	78.1	8.5	0.29

^a Density of grafted PEG molecules onto the PSZ chain expressed by the amount of PEG attaching to 10equiv of initial Si–H calculated according to $\delta = \frac{10}{Y} \times \left(1 - \frac{\%C_{\text{AP,iso}}}{100}\right)$.

where $I_{\text{CH}_2=\text{O}}$ is the integral of the $\text{CH}_2=$ resonance on the ^1H NMR spectrum of the initial mixture and $I_{\text{CH}_2=\text{O}(t)}$, $I_{\text{CH, cis}(t)}$ and $I_{\text{CH, trans}(t)}$ are the integrals of the $\text{CH}_2=$, cis CH= and trans CH= resonances at time t , respectively.

The olefin isomerization is accompanied by the appearance of a yellow color in the solution, which is generally ascribed to the irreversible formation of Pt colloid agglomerates [35]. It is now well established that oxygen may have a cocatalytic effect on hydrosilylation. One function of oxygen was to prevent and/or destruct the multinuclear Pt species. The simultaneous appearance of a yellow color with the isomerization products might suggest that alkene isomerization is also related to the presence of multinuclear platinum species. It is noted that a low amount of Pt catalyst ($\text{Pt/allyl} = 10^{-3}$) slowed down the reaction rate but did not prevent the formation of multinuclear Pt species as suggested by the yellow color of the grafted products. The comparison of results of argon and dry air atmosphere grafting (compare R_1 vs R_2) indicates that the isomerization reaction occurs more readily in the oxygen atmosphere than in the inert atmosphere. This suggests that the presence of oxygen does not destruct the multinuclear Pt species. The results in Table 1 (compare R_1 vs R_6 and R_3 vs R_5) also show that a high temperature promotes the isomerization reaction to the detriment of the hydrosilylation reaction. Besides, it is preferable to perform the reaction under solvent free conditions rather than in dilute medium (compare R_1 vs R_7). This is certainly due to the fact that dilution would favor the intramolecular isomerization reaction vs the intermolecular hydrosilylation reaction. From now on, the most favorable conditions for grafting allyl-PEG on PSZ were adopted: a temperature of 80°C with a Pt/allyl molar ratio of 3×10^{-3} under argon atmosphere and in solvent free conditions. The conversions of all three types of allyl-PEG in the different reactions during grafting as well as the density of grafted allyl-PEG molecules onto the PSZ chain are listed in Table 2. The kinetic curves obtained with allyl-PEG350 are shown in Fig. 2 as a representative example. It is noted that the reaction time to achieve a 100% conversion in allyl groups decreases and the degree of hydrosilylation increases when increasing the Si–H/allyl ratio and decreasing the molecular weight of PEG. Contrary to this, the degree of isomerization reduces significantly when increasing the Si–H/allyl ratio. In particular, when Si–H/allyl = 26.5, the *trans*-isomer could not be identified for both allyl-PEG750 and allyl-PEG2000 in their ^1H NMR spectra. A total conversion in allyl functionalities of MPEG750 and MPEG2000 is not reached after 55 h. As a general trend, the hydrosilylation reaction rate is slow. Electronic effects have been determined for the reactants in hydrosilylation reactions using Karstedt's catalyst: electron-donating groups on the olefin and electron-withdrawing groups on the hydrosilane increase the rate. There is strong

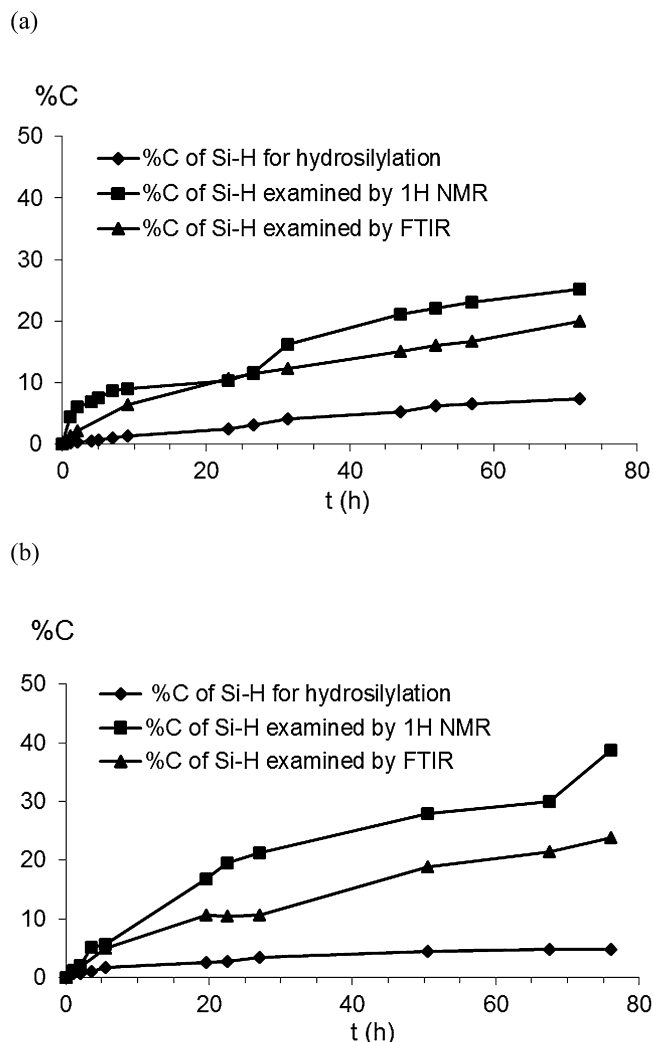


Fig. 3. % Conversion of SiH groups during grafting of allyl-PEG350 at $T = 80^\circ\text{C}$, $\text{Pt/allyl} = 0.001$ in a dry air atmosphere (a) and in argon atmosphere (b).

experimental evidence that the lone pair of the nitrogen is engaged in bonding with the silicon through the d-orbitals of the latter [36,37]. It is thus assumed that the entry of the π electrons of nitrogen into the d-orbitals of silicon would explain the low reactivity of SiH functional groups of polysilazane. The steric effects are also extremely important in hydrosilylation reaction [38]. The SEC chromatograms of the polysilazanes indicate a broad distribution of low molecular weights chains with a number-average molecular weight of 620 g/mol and a weight average molecular weight of 1000 g/mol. It is well-known that silazanes tend to form small ring cyclic products [39]. The FTIR spectra of PSZ does not show the two stretching bands near 3250 cm^{-1} and the deformation band near 1550 cm^{-1} , both characteristic of the terminal NH_2 functionalities. This suggests that cyclic structures predominate over linear oligomers. This might also explain the low reactivity of PSZ, by causing steric hindrance for the formation of the active Pt complexes.

The total consumption of Si–H groups were determined by two methods, the decrease of the Si–H peaks on ^1H NMR spectra centered at about 4.4 and 4.6 ppm and the decrease of the Si–H band at 2125 cm^{-1} in FTIR spectroscopy using the Si– CH_3 band at 1253 cm^{-1} as an internal reference (Fig. S3 in supporting information). The total conversion of Si–H groups determined by ^1H NMR and FTIR spectra (Fig. 3) was found to be much higher than the conversion associated with the hydrosilylation reaction (determined from the conversion of allyl-PEG for hydrosilylation, $\%C_{\text{AP,hydro}}$).

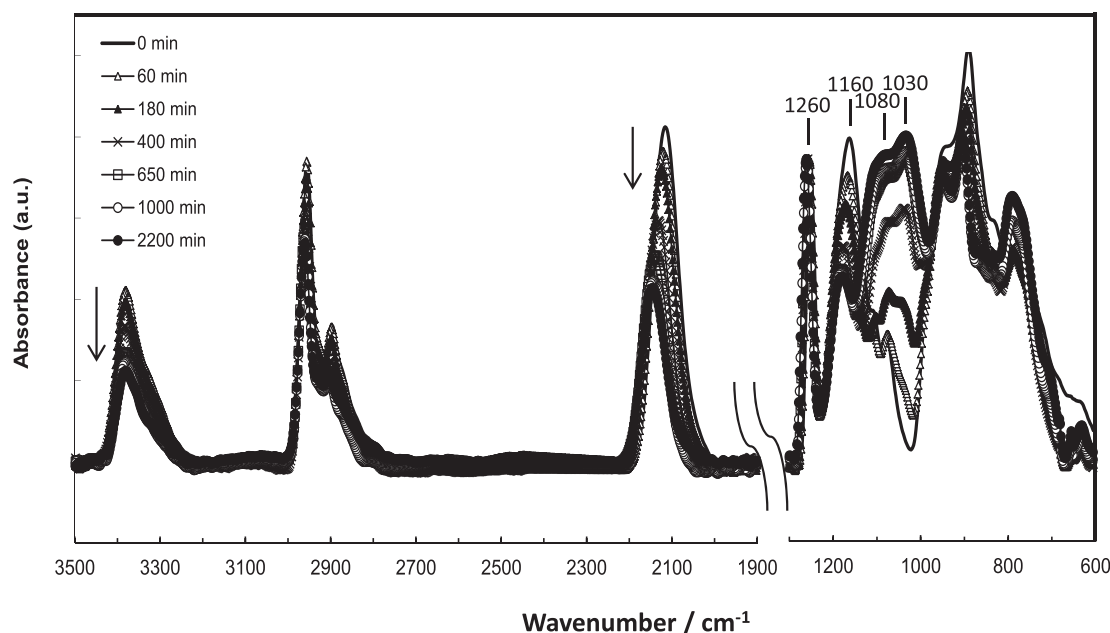
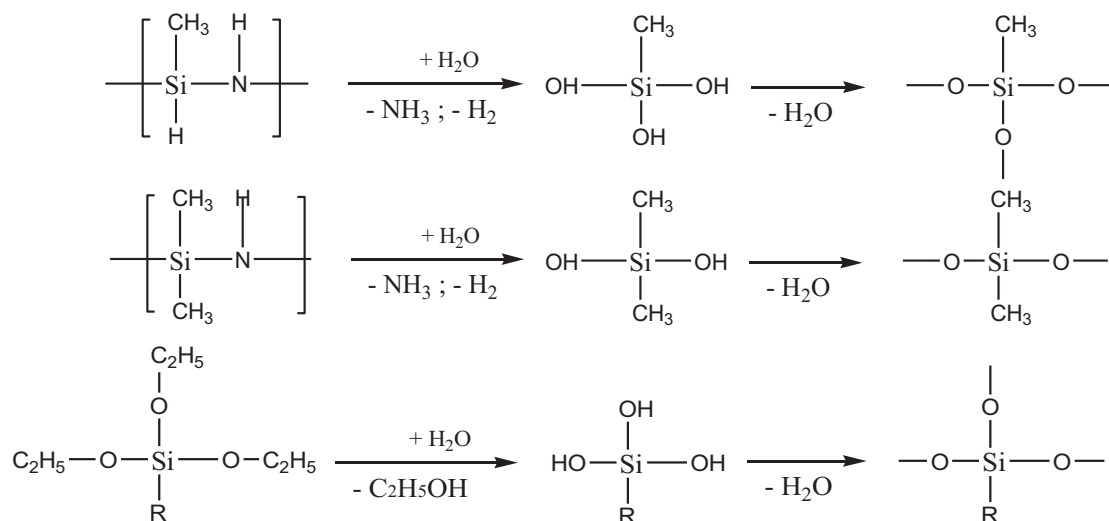


Fig. 4. FTIR spectra of PSZ after different curing times at 22 °C and 50% humidity.

It is suspected that SiH groups are consumed by the dehydrocoupling between SiH and NH bonds leading to a Si–N bond during the PEG grafting process. Further experiments suggested that dehydrocoupling of PSZ chain was activated by high temperature and Karstedt's catalyst. For example, a SiH conversion of ca. 5% was noted when PSZ was stirred for 5 days at 80 °C while the SiH conversion reaches ca. 30% when PSZ was stirred in the same conditions in presence of a small amount of Karstedt's catalyst. This agrees with the observation that dehydrocoupling is efficient in the presence of transition metal complexes [40]. Dehydrocoupling, in addition to transamination reactions [24], can explain the high viscosity of PSZ-PEG350s since these reactions result in an increased crosslink density. Despite the presence of the Pt catalyst residue in the PEG grafted PSZ products, the changes in SiH and NH functionalities were found to be insignificant during one month storage in ambient conditions. It is therefore possible to use the PEG grafted PSZ products several days after their implementation.

3.2. Solidification process in the ambient atmosphere of PSZ and PEG grafted PSZ

The fabrication of silazane derived ceramics involves cross-linking of the oligomeric silazane precursor followed by a thermal polymer-to-ceramic transformation step. Oligomeric silazanes are generally cross-linked either thermally or chemically using radical initiators like peroxides or catalysts. Alkoxy silyl substituted polysilazanes can be cross-linked at low temperature with the atmospheric moisture to form a solid film on the substrate surface. The changes in the infrared spectral patterns show that the bands at 3380 ($\nu(\text{NH})$), 2120 ($\nu(\text{SiH})$) and 1160 cm^{-1} (overlap between $\delta(\text{NH})$ [41] and $\rho(\text{CH}_3)$ of SiOEt groups [42]) all decrease with curing time (Fig. 4). Meanwhile, an increase of Si–O–Si vibrations at 1030 and 1080 cm^{-1} was observed. These results clearly indicate that the cure mechanism is based on the hydrolysis of Si–H, Si–N and alkoxy silyl bonds by water vapor to promote subsequent



Scheme 4. Reactions involved in the solidification of PSZ under ambient conditions.

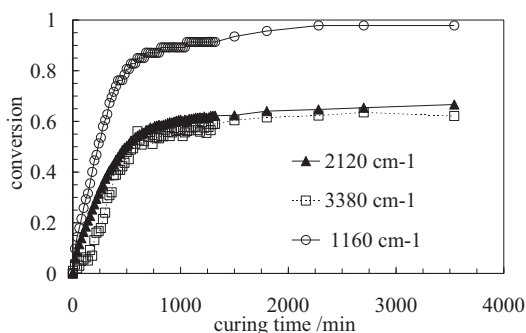


Fig. 5. Evolution of the 2120 cm⁻¹, 3380 cm⁻¹ and 1160 cm⁻¹ bands of PSZ with curing time at 22 °C and 50% humidity.

cross-linking of intermediately formed silanols via condensation reactions (Scheme 4).

The rate of disappearance of the 2120 cm⁻¹, 3380 cm⁻¹ and 1160 cm⁻¹ bands relative to the standard reference band at 1260 cm⁻¹, assigned to non-hydrolyzable SiCH₃ group, was used to quantitatively monitor the reaction. The conversion plots versus curing time shown in Fig. 5 reveal that the kinetics of SiH and NH transformation is nearly the same. Si–H and Si–N bonds thus possess the same reactivity toward OH groups, as noted by others [43]. Moreover, the intensity of the 1160 cm⁻¹ band appears to decrease much faster than the ν (SiH) and ν (NH) bands. Since the 1160 cm⁻¹ band is an overlap between the NH deformation and CH₃ deformation of Si(OEt) groups, it cannot be related strictly to the conversion of alkoxyisilyl groups. However, the fastest changes observed reveal that the hydrolysis of Si(OEt) groups is faster than the hydrolysis of Si–H and Si–N bonds. Ammonia which is released during curing brings an alkaline environment which can catalyze the hydrolysis (and self-condensation) of alkoxyisilyl groups (note that a γ -aminopropyltriethoxysilane film deposited from a sol at the natural pH ($\sim$ 11) reaches chemical equilibrium in a couple of minutes at room temperature [42]). It is also noted that as much as 40% of SiH and NH groups still remain in the film cured under ambient conditions (22 °C and a relative humidity of 50%). Fig. 6a shows that the grafting density of PEG350 has no significant effect on the moisture cure transformation of PSZ. Two antagonist effects are expected from the grafting of PEG molecules relative to the moisture cure kinetics of PSZ: an increased hydrophilicity which should increase the rate and steric crowding which should decrease it. The above result suggests that these two effects are balanced for PEG350 grafts. Fig. 6b shows the influence of the length of the PEG graft chains on the cure kinetics of PSZ. Compared to the PEG350 grafted PSZ, the rate of cure is significantly decreased with the PEG750 grafted PSZ. This suggests that steric crowding predominates over the hydrophilic effect for the long PEG grafts. However, the shape of the conversion curves in Fig. 6b suggests that the crosslinked density reached in the final equilibrium state is relatively unaffected by the length of the PEG graft chains. Table 3 shows the elemental composition of PSZ and PSZ–PEG350-7 uncured and moisture cured. The reaction of PSZ and PSZ–PEG-350-7 with moisture results in a decrease in N/Si, H/Si and C/Si ratio

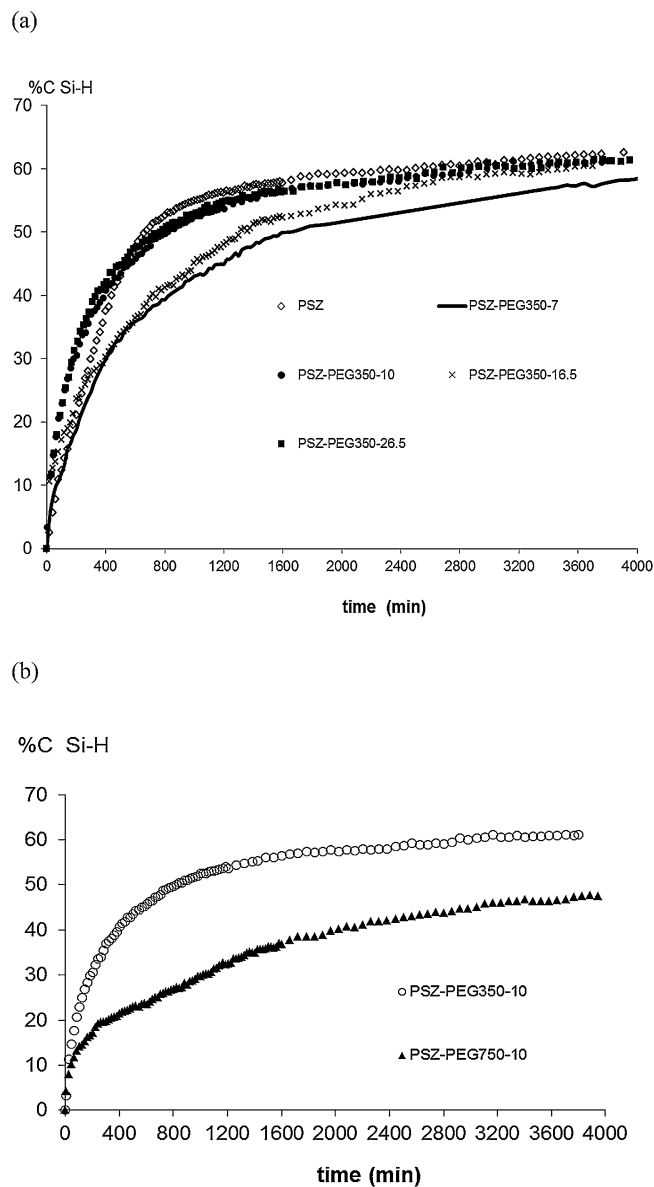


Fig. 6. SiH conversion as a function of curing time at 22 °C and 50% humidity (a) for PSZ and PEG350s-grafted PSZ and (b) for PEG350 and PEG750 grafted PSZs both at SiH/allyl ratio = 10.

which can be associated with the release of ammonia, hydrogen and ethanol. Meanwhile, the incorporation of oxygen suggested by the increase in O/Si ratio is associated with hydrolysis reactions on PSZ chains. The elemental analysis results thus confirm the moisture cure mechanism proposed in Scheme 4, which is common to PSZ and PEG grafted PSZ. The ²⁹Si NMR gives useful information on the coordination at the silicon atoms of the polysilazane chains after moisture curing. The ²⁹Si NMR spectrum of starting silazane

Table 3
Elemental analysis composition of PSZ and PSZ–PEG350-7 both uncured and cured (25 °C, 84% humidity).

Sample	Weight %					Atomic ratio X/Si				
	Si	C	H	N	O	Si	C	H	N	O
PSZ	31.84	27.21	8.66	18.97	13.33	1	1.99	7.62	1.19	0.73
PSZ (cured)	38.19	17.68	6.04	4.40	33.69	1	1.08	4.43	0.23	1.54
PSZ–PEG350-7	26.41	34.59	8.71	13.53	16.76	1	3.06	9.23	1.02	1.11
PSZ–PEG350-7 (cured)	25.40	30.59	6.80	3.34	33.87	1	2.81	7.50	0.26	2.33

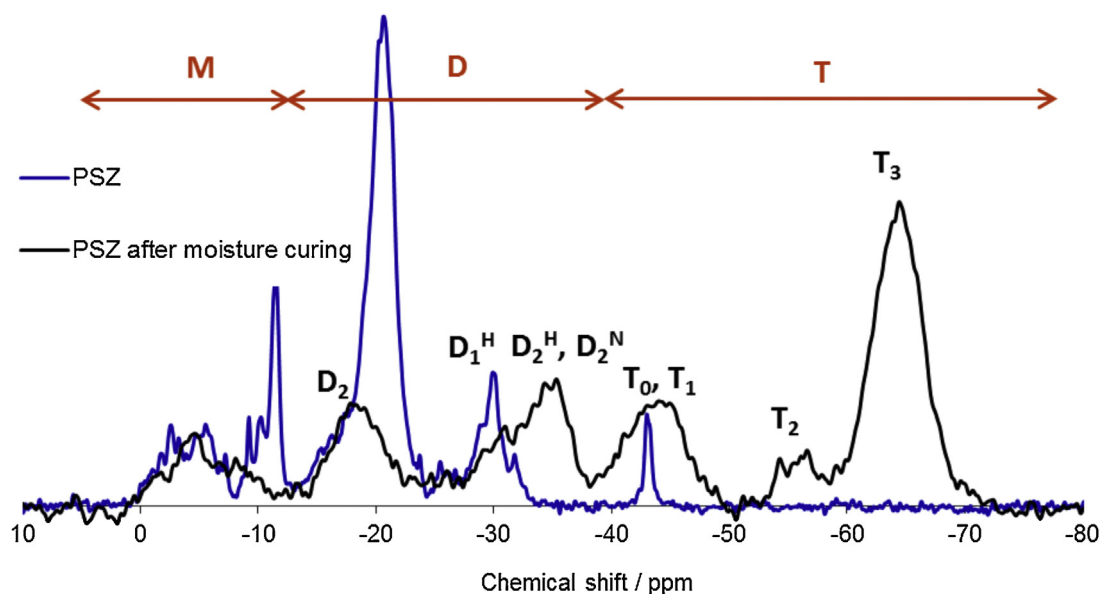


Fig. 7. ^{29}Si MAS NMR spectrum of PSZ and ^{29}Si CP/MAS NMR spectrum of the cross-linked polysilazane after moisture curing.

display three signal groups, one in the range from 2 to –13 ppm, corresponding to $(\text{CH}_3)_2\text{SiN}_2$ units [44–46], a second in the range from –13 to –35 ppm, corresponding to $\text{H}(\text{CH}_3)\text{SiN}_2$ units [46] and a broad peak around –43 ppm, corresponding to trialkoxysilyl units [44] (Fig. 7). The broad range of chemical shifts for the silazane units may be at least partly related to the cyclic structure of polysilazane with different ring sizes [46]. On the basis of chemical shift data from literature [44–47], the chemical shift ranges for the different types of silicon atoms in the moisture cured PSZ are >2.5 ppm for M_0^{N} , $+2.5$ ppm for M_1^{N} , 0 to –5 ppm for $\text{M}_0^{\text{H,N}}$, –12.5 ppm for $\text{M}_1^{\text{H,N}}$, –26 ppm for $\text{M}_1^{\text{N,N}}$, –12 to –13 ppm for D_1 , –16 to –25 ppm for D_2 , –27 ppm for D_1^{H} , –34 to –39 ppm for D_2^{H} , –33 ppm for D_1^{N} , –37 to –45 ppm for D_2^{N} , –40 to –48 ppm for T_0 , –45 to –50 ppm for T_1 , –52 to –60 ppm for T_2 and –61 to –70 ppm for T_3 .

*Calculated values assuming that changing from OH to OSi group produced a shift to higher field of 4–12 ppm as determined comparing the chemical shifts of D_1 and D_2 units and also, T_1 and T_2 units. **Calculated values based on the shielding effect of 15–25 ppm observed changing from Me to H [47].

The grafting of PEG chains on PSZ, new M and D units must be considered compared to the ungrafted PSZ: D_1^{PEG} , D_2^{PEG} , $\text{M}_0^{\text{PEG,N}}$ and $\text{M}_1^{\text{PEG,N}}$. However, the chemical shifts for the PEG substituted Si atom in the PEG grafted polysilazane must be close to those for their methyl homologs. The ^{29}Si solid-state NMR spectrum of PSZ after moisture curing displays significant differences compared to the spectrum of the PSZ precursor (Fig. 7). The cured PSZ exhibit the presence of various fragments based on T_0 , T_1 , T_2 and T_3 units, with the latter being the most intense. This confirms that a cross-linked siloxane network develops when PSZ is exposed to atmospheric moisture. It is also apparent that new linear structures after total (D_2) or partial (D_2^{H} and/or D_2^{N}) hydrolysis are built up. Grafting PEG onto the PSZ chains changes hydrolysable SiH groups to non-hydrolysable SiCH_2 groups. The PEG substituted Si atom cannot become a T unit but only a D_2 unit by hydrolysis-condensation reactions. Thus, the ratio between the intensity of the signals $(\text{D}_2 + \text{D}_2^{\text{PEG}})/(\text{T}_0 + \text{T}_1 + \text{T}_2 + \text{T}_3)$ should increase with the density of grafted PEG molecules. Such a trend is observed indeed (Table 4).

DSC analyses were used to follow the phase transition characteristics of PEG in MPEG, allyl-PEG and PEG grafted polysilazane and the results are shown in Fig. 8 and Table 5. From the DSC data,

Table 4

A comparison between grafting density and relative intensity between the D_2 and T signals as determined from ^{29}Si NMR spectra of the cross-linked PSZ and PSZ-PEG.

Sample	Molar % grafted PEG	$(\text{D}_2 + \text{D}_2^{\text{PEG}})/(\text{T}_0 + \text{T}_1 + \text{T}_2 + \text{T}_3)$
PSZ	0	0.24
PSZ-PEG2000-26.5	2.95	0.25
PSZ-PEG750-26.5	3.35	0.32
PSZ-PEG350-26.5	3.47	0.39
PSZ-PEG350-7	7.50	0.54

the degree of crystallinity X_c (%) was calculated using the following formula

$$X_c = \frac{\Delta H_m}{x_{\text{PEG}} \Delta H_m^\circ} \times 100$$

where ΔH_m° is the heat of melting of 100% crystalline polymer (196.8 J/g [48]), ΔH_m and X_c are the heat of melting and the weight fraction of allyl-PEG of the sample under investigation, respectively.

First, it was found that the change of a hydroxyl terminal group with allyl ether functionality has no significant effect on the thermal transitions of PEG chains. On the other hand, it was observed that an increase in the molecular weight of PEG chains in MPEG and allyl-PEG causes an increased degree of crystallinity, which agrees with previous observations [48]. This was ascribed to the lower segmental mobility and more convenient geometrical alignment of the higher molecular weight polymers. A glass-rubber transition was detected only for the lowest molecular weight (and crystallized) PEGs (MPEG 350 and allyl PEG 350). The grafting of PEG chains onto PSZ was found to strongly affect the thermal transitions of PEG. The more dramatic changes were observed with the lowest molecular weight PEG chains. Indeed, the stiff PSZ network totally inhibits the crystallization of PEG 350, while it only decreases the crystallization degree of PEG 750 and, in a more limited way, of PEG 2000. An additional effect observed for PEG 750 graft PSZ is associated with larger supercooling compared to MPEG 750 and allyl-PEG 750. The movement of PEG molecules is restrained and hindered in PEG grafted PSZ which destroyed orderliness of molecular chains, and decrease PEG crystallinity. The thermal decomposition behavior of the polymers in nitrogen and air atmosphere was studied by TGA. The results are summarized in Table 6 whereas TGA profiles are shown in Figs. 9 and 10. Low ceramic yields (30.6%) were

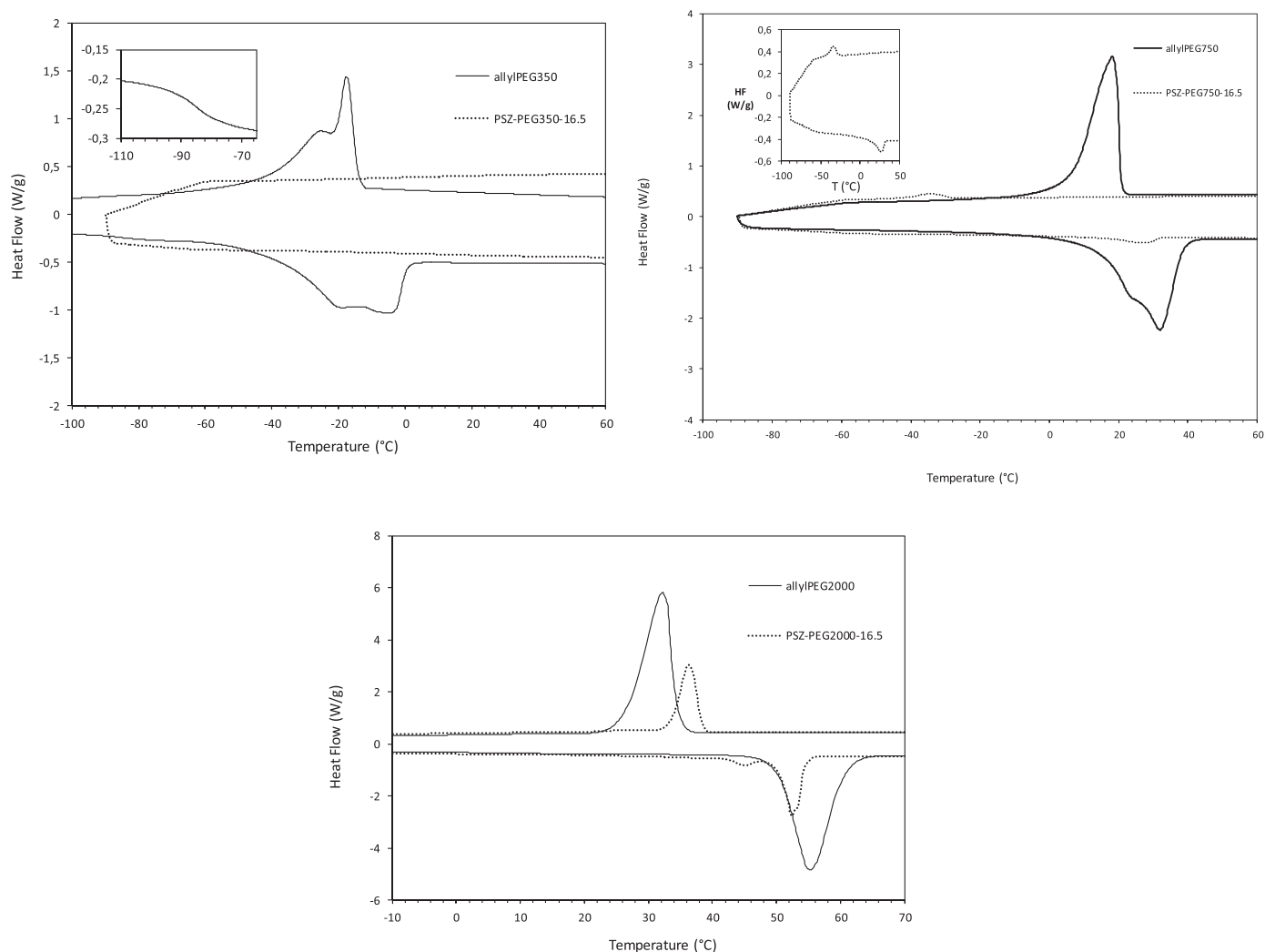


Fig. 8. DSC profiles for allyl-PEGs and PEG grafted PSZs.

obtained from the pyrolysis of the un-crosslinked oligosilazane in nitrogen (Fig. S4 in supporting information). This can be directly related to the fact that it does not contain thermally reactive functionalities such as the combination of SiH and vinyl groups in its structure. Indeed, high functionality is required to crosslink oligosilazane before it distills and, consequently, to achieve high ceramic yields [49]. The extensive weight loss observed at 50–500 °C can be

successively related to the distillation of the lightest oligomers and, from 150 °C to two kinds of depolymerization reactions: transamination reactions and bond redistribution around the silicon atom. Transamination is an exchange of N–Si and N–H bonds and thus involves the formation and scission of Si–N bonds. This may lead to the formation of volatile fragments and ammonia. The exchange between Si–N bonds and, to a lesser extent between Si–N and Si–H

Table 5
Thermal transition characteristics of PEG samples.

Sample	T_g (°C)	T_m (°C)	ΔH_m (J/g)	T_c (°C)	ΔH_c (J/g)	X_c (%)	$T_m - T_c$ (°C)
PSZ	–	–	–	–	–	–	–
MPEG350	–93.1	–3.4	109.3	–24.7	94.1	55.5	21.3
Allyl-PEG350	–85.7	–7.6	104.7	–17.7	104.8	53.2	10.1
^a PSZ-PEG350-X	–	–	–	–	–	–	–
MPEG750	–	31.0	163.3	18.1	161.8	83.0	12.9
Allyl-PEG750	–	31.9	160.6	18.2	162.1	81.6	13.7
PSZ-PEG750-10	–	12.9	4.6	–40.4	2.7	8.0	53.3
PSZ-PEG750-16.5	–	27.3	9.4	–35.2	6.0	21.6	60.4
PSZ-PEG750-26.5	–	26.0	7.1	–36.8	4.3	20.6	62.8
MPEG2000	–	55.2	193.7	33.2	170.5	98.4	22.0
Allyl-PEG2000	–	55.2	188.2	32.3	184.2	95.6	22.9
PSZ-PEG2000-10	–	53.0	65.8	35.0	58.4	66.1	18.0
PSZ-PEG2000-16.5	–	52.5	62.6	36.4	56.6	81.4	16.1
PSZ-PEG2000-26.5	–	51.9	59.4	36.7	58.3	90.6	15.2

– Not detected or not determined.

^a X = 7, 10, 16.5 or 26.5.

Table 6
TGA data of the polymers.

Sample	Nitrogen atmosphere			Air atmosphere		
	$T_{5\%}$ (°C) ^a	Weight loss (%)		$T_{5\%}$ (°C) ^a	Weight loss (%)	
		Exp	Calc ^b		Exp	Calc ^b
Cured PSZ	494	13.7	–	427	14.7	–
Allyl-PEG350	206	100	–	198	100	–
PSZ-PEG350-7	335	25.1	30.7	226	27.1	31.5
PSZ-PEG350-10	344	23.3	30.1	258	24.7	30.9
PSZ-PEG350-16.5	349	21.0	26.0	277	22.9	26.8
PSZ-PEG350-26.5	372	19.2	22.3	333	20.7	23.2
Allyl-PEG750	239	100	–	227	100	–
PSZ-PEG750-10	319	32.5	38.8	202	34.1	39.8
PSZ-PEG750-16.5	315	29.9	32.5	205	32.7	33.6
PSZ-PEG750-26.5	334	24.6	28.5	213	29.1	29.6
Allyl-PEG2000	342	100	–	203	100	–
PSZ-PEG2000-10	293	55.6	57.7	188	58.0	57.9
PSZ-PEG2000-16.5	295	48.3	57.4	188	51.8	48.1
PSZ-PEG2000-26.5	288	44.0	42.6	186	47.7	43.2

^a Temperature at 5% weight loss.

^b Calculated weight loss (in %) considering that PEG chains are totally lost and that the weight loss for PSZ in PEG grafted PSZ is the same than in the cured ungrafted PSZ.

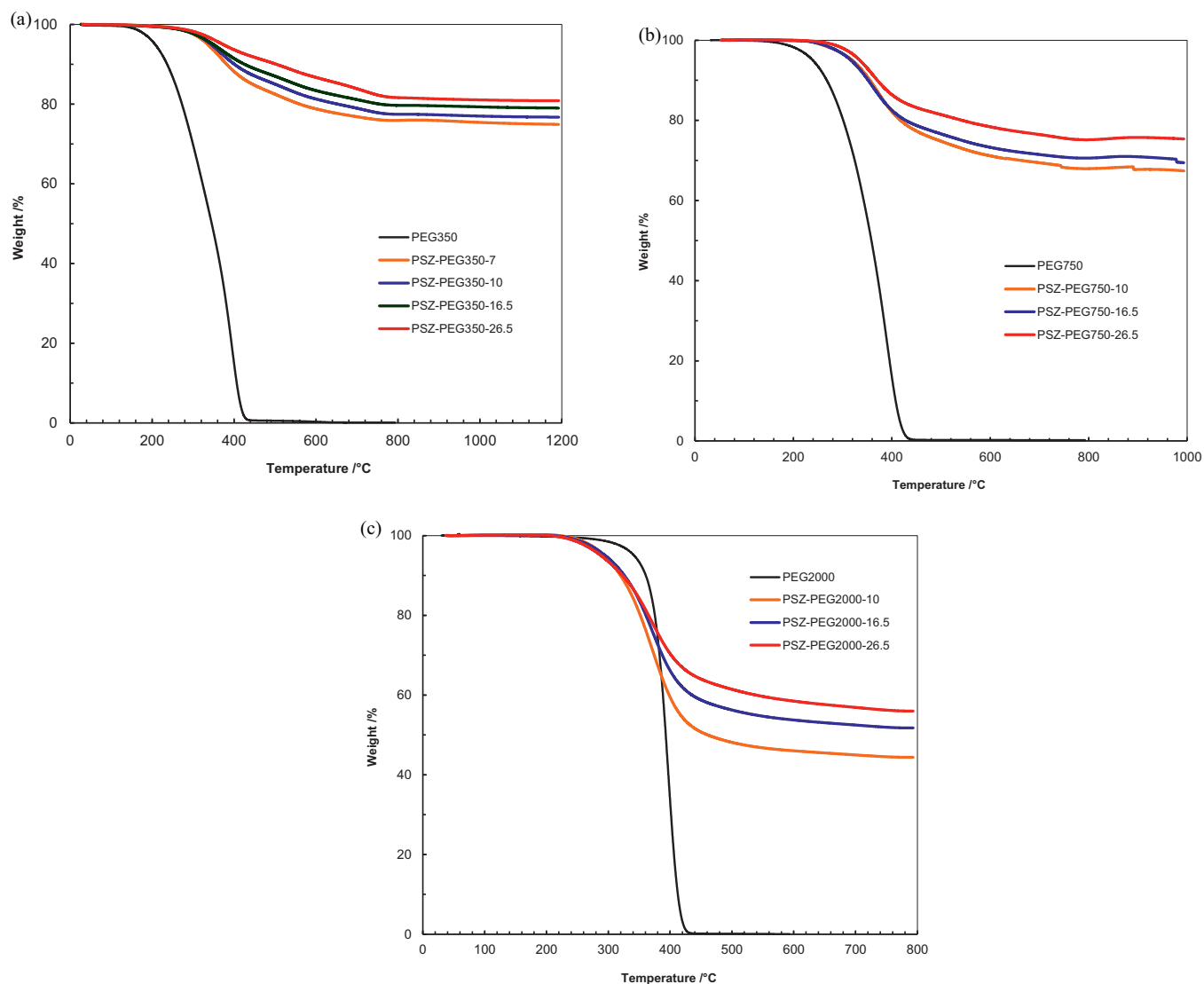


Fig. 9. TGA profiles of allyl-PEG and PEG grafted PSZs from (a) PEG350, (b) PEG750 and (c) PEG2000 under nitrogen.

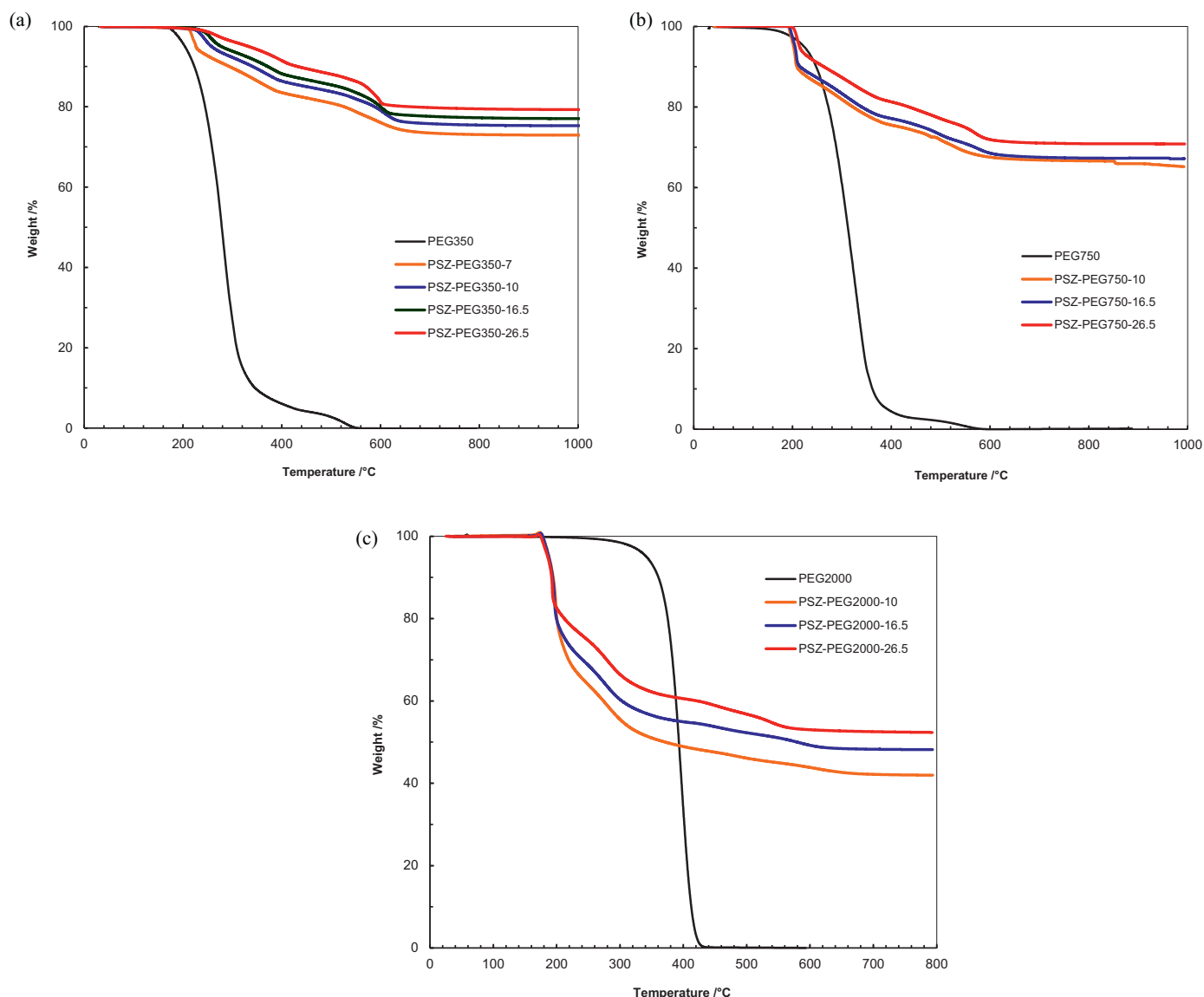


Fig. 10. TGA profiles of allyl-PEG and PEG grafted PSZs from (a) PEG350, (b) PEG750 and (c) PEG2000 under air.

bonds may also lead to the formation of volatile oligomers and silane. Dehydrogenation reactions between SiH and NH also occur in the same temperature range [49]. Above 500 °C, a second weight loss (9%) corresponds to the mineralization step and mainly the release of methane and hydrogen [50]. The formation of methane arises from radical cleavage of SiCH₃ followed by H abstraction (from SiH, NH or CH). The formation of hydrogen is due to radical cleavage of SiH, NH and CH bonds followed by H abstraction [51]. For the moisture cured silazane, the pyrolysis behavior changed dramatically (Fig. S4). A ceramic yield of 86.3% was achieved and the decomposition starts only above 350 °C. The moisture curing results in a high degree of crosslinking which limits the distillation of oligosilazane. Besides, transamination reactions and the exchange between SiH and SiN bonds are certainly less operative in the cured silazane due to the decrease in SiH and NH functionalities resulting from the moisture curing. Under an oxidative atmosphere, a higher ceramic yield (54.4%) was obtained for the uncrosslinked silazane (Fig. S4). This is due to the incorporation of oxygen in the product via oxidation of SiH and SiN bonds followed by three kinds of crosslinked reactions to form siloxane bonds: reactions between two SiOHs, between SiOH and SiH and between SiOH and

SiN [51]. The white ceramic residue at 1000 °C in air is silica which shows that the combustion of hydrocarbon groups occurred. It is noteworthy that the built up of the siloxane network competes with the reactions described before in a nitrogen environment as the ceramic yield obtained in air is far lower than the theoretical ceramic yield corresponding to a total conversion of PSZ to silica (12.4%). The ceramic yield for the crosslinked PSZ pyrolyzed in air is nearly the same than in nitrogen atmosphere. For this polymer, oxygen has been already incorporated during the moisture curing which results in small differences of thermal behavior between the two environments. Interestingly, the ceramic residue contained a significant amount of carbon as revealed by its gray color. The thermal decomposition of allyl-PEG 2000 under a nitrogen atmosphere proceeds through one main step at ca. 400 °C which agrees with previously published results on PEG with a molecular weight around 10,000 g/mol [48] (Fig. 9). The thermal decomposition of allyl PEG 350 and allyl PEG 750 extends over a broader temperature range which may be due to the polydispersity of these polymers. The weight loss curves show that the thermal stability of allyl-PEG in a non-oxidative atmosphere increases with the molecular weight of the PEG chains while no clear trend is observed in air (Fig. 10),

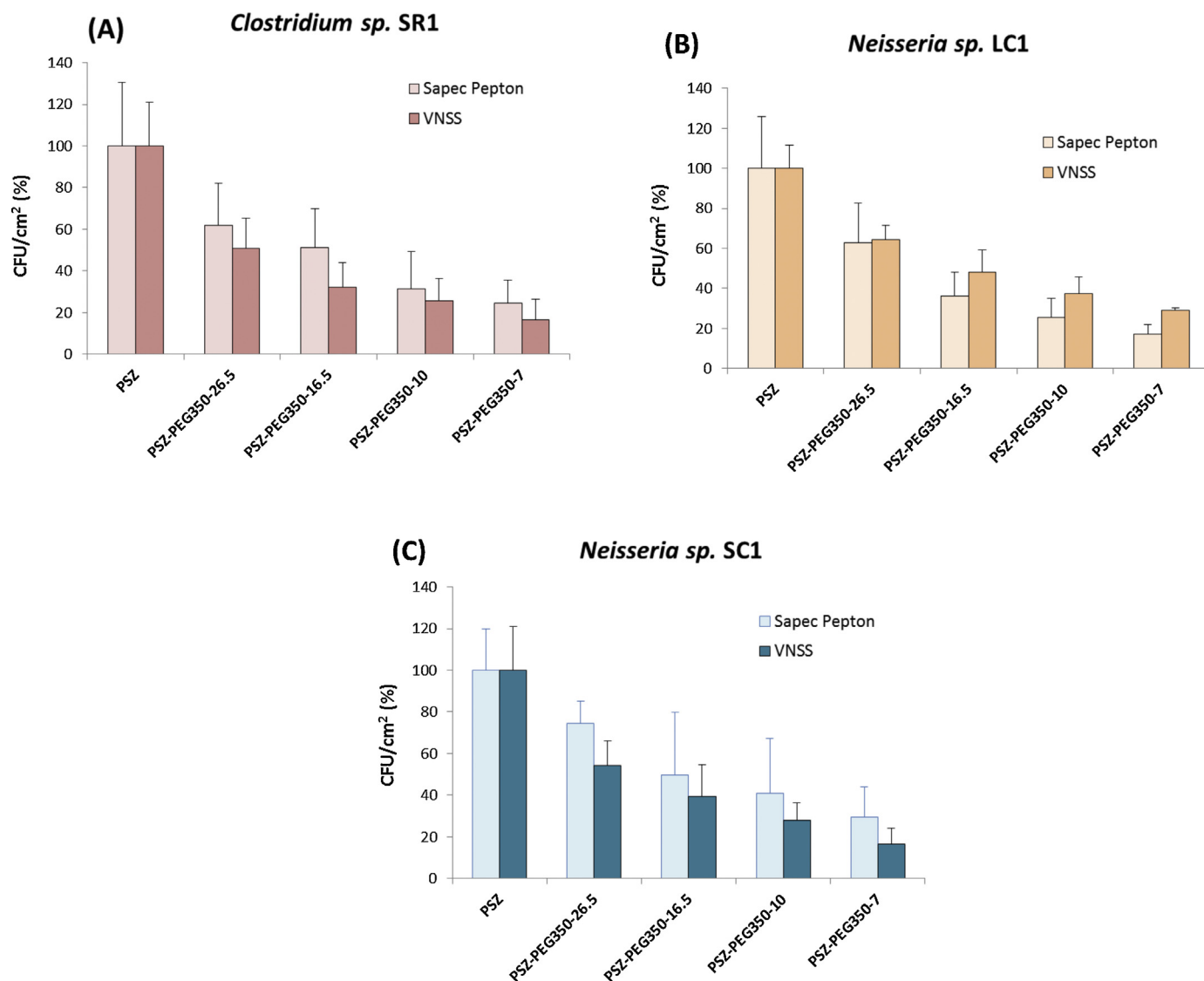


Fig. 11. Adhesion of *Clostridium* sp. SR1 (a), *Neisseria* sp. LC1 (b) and *Neisseria* sp. SC1 (c) on the PSZ and PSZ-PEG350s coatings. Error bars represent the average standard deviation over three separate experiments.

allyl PEG 750 appearing as the most stable. As expected, the thermal stability of PSZ-PEGs is lower than that of PSZ due to the low stability of PEG chains compared to PSZ. The temperature at 5% weight loss generally increases with the ratio SiH:allyl, and thus, when the PEG amount decreases in PSZ-PEGs. The PEG2000 (under nitrogen and air) and PEG750 (under air) chains appear less stable in PSZ-PEG compared to allyl-PEG which could be due to the lower crystallinity of PEG chains in the former case. The hydrosilylation reaction of allyl-PEG changes SiH functionalities to SiCH₂ linking which must result in a lower loss for PSZ in PSZ-PEGs by dehydrogenation between SiH and NH and by exchange reactions between SiH and SiN. This would favor the formation of excess carbon in the residue and explain the fact that ceramic yields generally appear higher than expected in theory (Table 6).

3.3. Marine bacterial adhesion on modified surfaces

The resistance of the PSZ surfaces to the adhesion of three marine bacteria species, bacillus *Clostridium* sp. SR1, micrococcus *Neisseria* sp. LC1 and micrococcus *Neisseria* sp. SC1 was investigated. Obtained results were interpreted according to the influence of the Si–H/allyl ratio (Fig. 11) and of the length of the grafted PEG

(Fig. 12). Figs. 11 and 12 show that all PEG-g-PSZ surfaces exhibit lower levels of bacteria adhesion compared to the unmodified PSZ coating, irrespective of the culture media used for incubation. The similar behavior of different bacteria on the same surface suggests that the adsorption of bacteria is dominated by the interfacial properties of the PEG-g-PSZ surface not by those of the bacteria. The ability to inhibit the bacterial adhesion increases with the decrease of Si–H/Allyl molar ratio and therefore, with the increase of PEG grafts density. Thus, the PSZ PEG350-7 surface has the best inhibitory properties and leads to a reduction of the number of adhered bacteria approximately 80% relative to the PSZ coating without PEG grafts. In comparison, PSZ-PEG350-26.5 surface results in a reduction of approximately 40% in the number of adhered bacteria. The influence of the length of PEG grafts on marine bacterial adhesion was investigated with the Si–H/Allyl ratio of 10 in VNSS and Saptec-pepton media and shown in Fig. 12a and b, respectively. PSZ-PEG350 coating was the most inhibitive surface. There are large discrepancies in the data available on the effect of PEG molecular weight on the efficiency in PEG coatings for reducing bacterial and protein adsorption. Several researchers have shown that long PEG chains resist protein and cell adhesion more effectively than short chains [4,9,52–54]. However, Prime

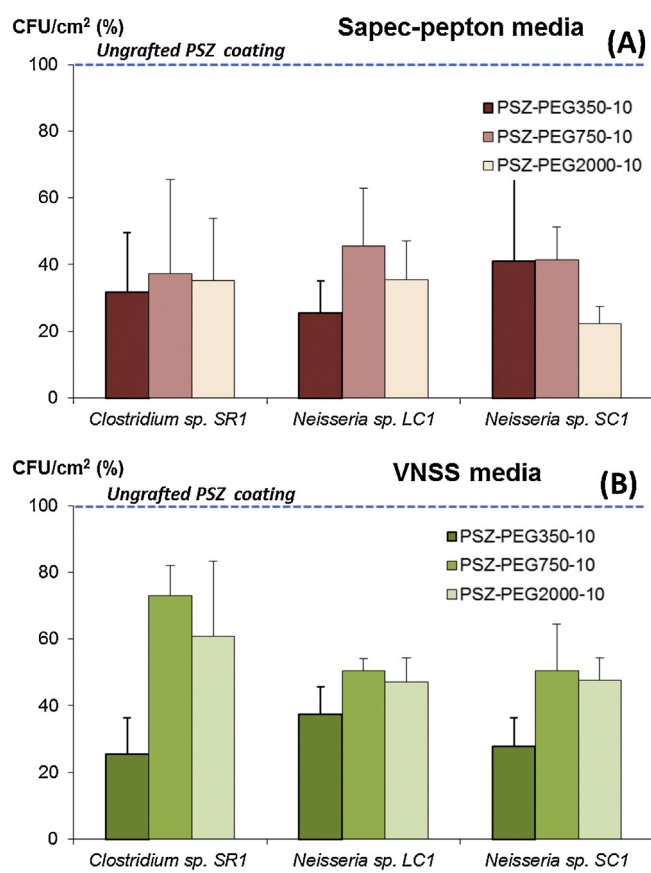


Fig. 12. Evolution of the adhesion of three marine bacteria for different lengths of PEG chain in Sapec-pepton (a) and VNSS (b) media at Si–H/allyl molar ratio of 10.

et al. [30] suggested that dense oligoethylene oxide layers with only three ethylene oxide segments prevented protein adsorption as effectively as long chains. Based on a surface–protein interaction model Jeon et al. [55] demonstrated that the surface density of the PEG chains have a greater effect than the length of the PEG chains in inhibiting bacterial adhesion. All PSZ–PEG350 coatings have a PEG grafting density that is higher than that of PSZ–PEG750 and PSZ–PEG2000 at the same Si–H/allyl ratio (Table 2). The higher PEG grafting density of PSZ–PEG350 could explain its higher antifouling activity. As the grafting density of PEG750 and PEG2000 onto PSZ is approximately equivalent, the effectiveness of these two types of hybrid materials is controlled by the chain length of PEG. Consequently, the PSZ–PEG2000 surfaces are more efficient than the PSZ–PEG750 surfaces regarding to the antifouling activity.

4. Conclusions

The PSZs and PEG(350)-graft-PSZs are low viscous liquids which can be directly used for coating applications in low VOC solvent free formulations. The grafting of PEG chains onto polysilazane by hydrosilylation reaction and, thus, through robust silicon–carbon bonds produce an antifouling surface with long-term stability. The crystallization of short PEG chains (350 g/mol) was found to be totally inhibited whereas long PEG grafts (750 and 2000 g/mol) are still able to crystallize. The PSZ coatings led to excellent ceramic yield (over 86%) after moisture curing at room temperature. A high ceramic yield (>86%) was achieved upon pyrolysis for the moisture cured polysilazane due to its high crosslinked density which limits the distillation of oligosilazane. This behavior was ascribed to the high reactivity to moisture of oligosilazane chains, which allows

efficient cross-linking. The ceramic yields generally appear higher than expected in theory in the PEG-graft-PSZ coatings which was related to the lower loss by dehydrogenation between SiH and NH and by exchange reactions between SiH and SiN compared to the ungrafted PSZ. The better bacterial resistance of PEG350-grafted-PSZ compared to that of the PEG(750) and PEG(2000)-graft-PSZs can be attributed to the highest grafting density of the shortest PEG350. Another contributing factor could be the fact that the grafted PEG350 do not pack as crystals contrary to the grafted PEG750 and PEG2000 which must result in a larger entropy loss associated with bacteria adsorption on top of the loosely packed and more mobile PEG350-grafted-PSZ surface.

Acknowledgements

The authors acknowledge the Clariant Company for its kind supply of the polysilazane. We would like to thank Dr. Can Vu for fruitful discussions about silazane chemistry.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.porgcoat.2015.06.022>

References

- [1] D.M. Yebra, S. Kiil, K. Dam-Johansen, Antifouling technology—past, present and future steps towards efficient and environmentally friendly antifouling coatings, *Prog. Org. Coat.* 50 (2004) 75–104, <http://dx.doi.org/10.1016/j.porgcoat.2003.06.001>
- [2] S. Krishnan, C.J. Weinman, C.K. Ober, Advances in polymers for anti-biofouling surfaces, *J. Mater. Chem.* 18 (2008) 3405–3413, <http://dx.doi.org/10.1039/B801491D>
- [3] J.P. Youngblood, L. Andruzzi, C.K. Ober, A. Hexemer, E.J. Kramer, J.A. Callow, et al., Coatings based on side-chain ether-linked poly(ethylene glycol) and fluorocarbon polymers for the control of marine biofouling, *Biofouling* 19 (2003) 91–98, <http://dx.doi.org/10.1080/0892701021000053381>
- [4] A. Roosjen, H.C. van der Mei, H.J. Busscher, W. Norde, Microbial adhesion to poly(ethylene oxide) brushes: influence of polymer chain length and temperature, *Langmuir* 20 (2004) 10949–10955, <http://dx.doi.org/10.1021/la048469l>
- [5] B.S. Flavel, M. Jasieniak, L. Velleman, S. Ciampi, E. Luais, J.R. Peterson, et al., Grafting of poly(ethylene glycol) on click chemistry modified Si(100) surfaces, *Langmuir* 29 (2013) 8355–8362, <http://dx.doi.org/10.1021/la400721c>
- [6] M. Li, K.G. Neoh, L.Q. Xu, R. Wang, E.-T. Kang, T. Lau, et al., Surface modification of silicone for biomedical applications requiring long-term antibacterial, antifouling, and hemocompatible properties, *Langmuir* 28 (2012) 16408–16422, <http://dx.doi.org/10.1021/la303438t>
- [7] S. Li, D. Yang, H. Tu, H. Deng, D. Du, A. Zhang, Protein adsorption and cell adhesion controlled by the surface chemistry of binary perfluoroalkyl/oligo(ethylene glycol) self-assembled monolayers, *J. Colloid Interface Sci.* 402 (2013) 284–290, <http://dx.doi.org/10.1016/j.jcis.2013.04.003>
- [8] T. Ekblad, G. Bergström, T. Ederth, S.L. Conlan, R. Mutton, A.S. Clare, et al., Poly(ethylene glycol)-containing hydrogel surfaces for antifouling applications in marine and freshwater environments, *Biomacromolecules* 9 (2008) 2775–2783, <http://dx.doi.org/10.1021/bm800547m>
- [9] O. Iguerb, C. Poleunis, F. Mazéas, C. Compère, P. Bertrand, Antifouling properties of poly(methyl methacrylate) films grafted with poly(ethylene glycol) monoacrylate immersed in seawater, *Langmuir* 24 (2008) 12272–12281, <http://dx.doi.org/10.1021/la801814u>
- [10] G. Kang, H. Yu, Z. Liu, Y. Cao, Surface modification of a commercial thin film composite polyamide reverse osmosis membrane by carbodiimide-induced grafting with poly(ethylene glycol) derivatives, *Desalination* 275 (2011) 252–259, <http://dx.doi.org/10.1016/j.desal.2011.03.007>
- [11] Y.-L. Su, W. Cheng, C. Li, Z. Jiang, Preparation of antifouling ultrafiltration membranes with poly(ethylene glycol)-graft-polyacrylonitrile copolymers, *J. Membr. Sci.* 329 (2009) 246–252, <http://dx.doi.org/10.1016/j.memsci.2009.01.002>
- [12] I.M. Nnebe, R.D. Tilton, J.W. Schneider, Direct force measurement of the stability of poly(ethylene glycol)–polyethylenimine graft films, *J. Colloid Interface Sci.* 276 (2004) 306–316, <http://dx.doi.org/10.1016/j.jcis.2004.03.065>
- [13] C. Choi, I. Hwang, Y.-L. Cho, S.Y. Han, D.H. Jo, D. Jung, et al., Fabrication and characterization of plasma-polymerized poly(ethylene glycol) film with

- superior biocompatibility, *ACS Appl. Mater. Interfaces* 5 (2013) 697–702, <http://dx.doi.org/10.1021/am302208f>
- [14] P. Kingshott, J. Wei, D. Bagge-Ravn, N. Gadegaard, L. Gram, Covalent attachment of poly(ethylene glycol) to surfaces, critical for reducing bacterial adhesion, *Langmuir* 19 (2003) 6912–6921, <http://dx.doi.org/10.1021/la034032m>
 - [15] G. Kang, Y. Cao, H. Zhao, Q. Yuan, Preparation and characterization of crosslinked poly(ethylene glycol) diacrylate membranes with excellent antifouling and solvent-resistant properties, *J. Membr. Sci.* 318 (2008) 227–232, <http://dx.doi.org/10.1016/j.memsci.2008.02.045>
 - [16] J. Zhou, J. Yin, R. Lv, Q. Du, W. Zhong, Preparation and properties of MPEG-grafted EAA membranes via thermally induced phase separation, *J. Membr. Sci.* 267 (2005) 90–98, <http://dx.doi.org/10.1016/j.memsci.2005.06.002>
 - [17] C.S. Gudipati, J.A. Finlay, J.A. Callow, M.E. Callow, K.L. Wooley, The antifouling and fouling-release performance of hyperbranched fluoropolymer (HBFP)-poly(ethylene glycol) (PEG) composite coatings evaluated by adsorption of biomacromolecules and the green fouling alga *Ulva*, *Langmuir* 21 (2005) 3044–3053, <http://dx.doi.org/10.1021/la048015a>
 - [18] A. Razatos, Y.-L. Ong, F. Boulay, D.L. Elbert, J.A. Hubbell, M.M. Sharma, et al., Force measurements between bacteria and poly(ethylene glycol)-coated surfaces, *Langmuir* 16 (2000) 9155–9158, <http://dx.doi.org/10.1021/la000818y>
 - [19] I.C. Saldarriaga Fernández, H.C. van der Mei, M.J. Lochhead, D.W. Grainger, H.J. Busscher, The inhibition of the adhesion of clinically isolated bacterial strains on multi-component cross-linked poly(ethylene glycol)-based polymer coatings, *Biomaterials* 28 (2007) 4105–4112, <http://dx.doi.org/10.1016/j.biomaterials.2007.05.023>
 - [20] J. Peyre, V. Humblot, C. Méthivier, J.-M. Berjeaud, C.-M. Pradier, Co-grafting of amino-poly(ethylene glycol) and Magainin I on a TiO₂ surface: tests of antifouling and antibacterial activities, *J. Phys. Chem. B* 116 (2012) 13839–13847, <http://dx.doi.org/10.1021/jp305597y>
 - [21] Q. Wang, E. Uzunoglu, Y. Wu, M. Libera, Self-assembled poly(ethylene glycol)-co-acrylic acid microgels to inhibit bacterial colonization of synthetic surfaces, *ACS Appl. Mater. Interfaces* 4 (2012) 2498–2506, <http://dx.doi.org/10.1021/am300197m>
 - [22] Y. Cho, H.S. Sundaram, C.J. Weinman, M.Y. Paik, M.D. Dimitriou, J.A. Finlay, et al., Triblock copolymers with grafted fluorine-free, amphiphilic, non-ionic side chains for antifouling and fouling-release applications, *Macromolecules* 44 (2011) 4783–4792, <http://dx.doi.org/10.1021/ma200269s>
 - [23] D. Park, C.J. Weinman, J.A. Finlay, B.R. Fletcher, M.Y. Paik, H.S. Sundaram, et al., Amphiphilic surface active triblock copolymers with mixed hydrophobic and hydrophilic side chains for tuned marine fouling-release properties, *Langmuir* 26 (2010) 9772–9781, <http://dx.doi.org/10.1021/la100032n>
 - [24] E. Kroke, Y.-L. Li, C. Konetschny, E. Lecomte, C. Fasel, R. Riedel, Silazane derived ceramics and related materials, *Mater. Sci. Eng. R: Rep.* 26 (2000) 97–199, [http://dx.doi.org/10.1016/S0927-796X\(00\)00008-5](http://dx.doi.org/10.1016/S0927-796X(00)00008-5)
 - [25] S. Brand, A. Dierdorf, H. Liebe, A. Wacker, Use of Polysilazanes as Permanent Anti-Fingerprint Coatings, WO/2006/089649, 2006. <https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2006089649> (accessed 12.06.15).
 - [26] R. Bhandavat, A. Feldman, C. Cromer, J. Lehman, G. Singh, Very high laser-damage threshold of polymer-derived Si(B)CN-carbon nanotube composite coatings, *ACS Appl. Mater. Interfaces* 5 (2013) 2354–2359, <http://dx.doi.org/10.1021/am302755x>
 - [27] L. David, D. Asok, G. Singh, Synthesis and extreme rate capability of Si—Al—C—N functionalized carbon nanotube spray-on coatings as Li-ion battery electrode, *ACS Appl. Mater. Interfaces* 6 (2014) 16056–16064, <http://dx.doi.org/10.1021/am5052729>
 - [28] I.I.I. Lukacs, G.J. Knasiak, United States Patent: 6652978 – Thermally stable, moisture curable polysilazanes and polysiloxazanes, 6652978, 2003. <http://patft.uspto.gov/netacgi/nph-Parser?Sect1=PTO1&Sect2=HITOFF&d=PALL&p=1&u=%2Fnetacgi%2FPTO%2Fsrchnum.htm&r=1&f=G&l=50&s1=6,652,978.PN&OS=PN/6,652,978&RS=PN/6,652,978> (accessed 07.01.14).
 - [29] S. Marceaux, C. Bressy, F.-X. Perrin, C. Martin, A. Margaillan, Development of polyorganosilazane-silicone marine coatings, *Prog. Org. Coat.* 77 (2014) 1919–1928, <http://dx.doi.org/10.1016/j.porgcoat.2014.06.020>
 - [30] K.L. Prime, G.M. Whitesides, Adsorption of proteins onto surfaces containing end-attached oligo(ethylene oxide): a model system using self-assembled monolayers, *J. Am. Chem. Soc.* 115 (1993) 10714–10721, <http://dx.doi.org/10.1021/ja00076a032>
 - [31] T.D.H. Nguyen, F.-X. Perrin, D.L. Nguyen, New hybrid materials based on poly(ethyleneoxide)-grafted polysilazane by hydrosilylation and their anti-fouling activities, *Beilstein J. Nanotechnol.* 4 (2013) 671–677, <http://dx.doi.org/10.3762/bjnano.4.75>
 - [32] L. Lestel, H. Cheradame, S. Boileau, Crosslinking of polyether networks by hydrosilylation and related side reactions, *Polymer* 31 (1990) 1154–1158, [http://dx.doi.org/10.1016/0032-3861\(90\)90266-2](http://dx.doi.org/10.1016/0032-3861(90)90266-2)
 - [33] S.-S. Hou, Y.-P. Chung, C.-K. Chan, P.-L. Kuo, Function and performance of silicone copolymer. Part IV. Curing behavior and characterization of epoxy-siloxane copolymers blended with diglycidyl ether of bisphenol-A, *Polymer* 41 (2000) 3263–3272, [http://dx.doi.org/10.1016/S0032-3861\(99\)00525-X](http://dx.doi.org/10.1016/S0032-3861(99)00525-X)
 - [34] D.-W. Chung, T.G. Kim, Study on the effect of platinum catalyst for the synthesis of polydimethylsiloxane grafted with polyoxyethylene, *J. Ind. Eng. Chem.* 13 (2007) 571–577.
 - [35] L.N. Lewis, On the mechanism of metal colloid catalyzed hydrosilylation: proposed explanations for electronic effects and oxygen cocatalysis, *J. Am. Chem. Soc.* 112 (1990) 5998–6004, <http://dx.doi.org/10.1021/ja00172a014>
 - [36] K. Hedberg, The molecular structure of trisilylamine (SiH₃)₃N^{1,2}, *J. Am. Chem. Soc.* 77 (1955) 6491–6492, <http://dx.doi.org/10.1021/ja01629a015>
 - [37] A.G. MacDiarmid, A.G. Maddock, The structure of silyl iso-thiocyanate, *J. Inorg. Nucl. Chem.* 1 (1955) 411, [http://dx.doi.org/10.1016/0022-1902\(55\)80050-4](http://dx.doi.org/10.1016/0022-1902(55)80050-4)
 - [38] J. Stein, L.N. Lewis, K.A. Smith, K.X. Lettko, Mechanistic studies of platinum-catalyzed hydrosilylation, *J. Inorg. Organomet. Polym.* 1 (1991) 325–334, <http://dx.doi.org/10.1007/BF00702496>
 - [39] M. Birot, J.-P. Pillot, J. Dunogues, Comprehensive chemistry of polycarbosilanes, polysilazanes, and polycarbosilazanes as precursors of ceramics, *Chem. Rev.* 95 (1995) 1443–1477, <http://dx.doi.org/10.1021/cr00037a014>
 - [40] R. Laine, Y. Blum, A. Chow, R. Hamlin, K. Schwartz, D. Rowcliffe, Catalytic synthesis of novel polysilazanes including precursors to silicon nitride, *Polym. Prepr.* 28 (1987) 393–395.
 - [41] H.N. Han, D.A. Lindquist, J.S. Haggerty, D. Seyferth, Pyrolysis chemistry of poly(organosilazanes) to silicon ceramics, *Chem. Mater.* 4 (1992) 705–711, <http://dx.doi.org/10.1021/cm00021a038>
 - [42] V. Roche, F.X. Perrin, D. Gignes, F. Vacandio, F. Ziarelli, D. Bertin, Tracking the fate of γ -aminopropyltriethoxysilane from the sol state to the dried film state, *Thin Solid Films* 518 (2010) 3640–3645, <http://dx.doi.org/10.1016/j.tsf.2009.09.085>
 - [43] F. Bauer, U. Decker, A. Dierdorf, H. Ernst, R. Heller, H. Liebe, et al., Preparation of moisture curable polysilazane coatings: Part I. Elucidation of low temperature curing kinetics by FT-IR spectroscopy, *Prog. Org. Coat.* 53 (2005) 183–190, <http://dx.doi.org/10.1016/j.porgcoat.2005.02.006>
 - [44] S. Ek, E.I. Iiskola, L. Niinistö, J. Vaittinen, T.T. Pakkanen, A. Root, A ²⁹Si and ¹³C CP/MAS NMR study on the surface species of gas-phase-deposited γ -aminopropylalkoxysilanes on heat-treated silica, *J. Phys. Chem. B* 108 (2004) 11454–11463, <http://dx.doi.org/10.1021/jp048927z>
 - [45] H. Tang, R.E. Prud'homme, A.-F. Mingotaud, M. Schappacher, A. Soum, Solid-state characterization and properties of poly(N-methylcyclodisilazanes), *Macromolecules* 30 (1997) 1400–1407, <http://dx.doi.org/10.1021/ma961209t>
 - [46] N.S. Choong Kwet Yive, R. Corriu, D. Leclercq, P.H. Mutin, A. Vioux, Polyvinylsilazane: a novel precursor to silicon carbonitride, *New J. Chem.* 15 (1991) 85–92.
 - [47] E.A. Williams, NMR spectroscopy of organosilicon compounds, in: S. Patai, Z. Rappoport (Eds.), *Chem. Funct. Groups*, John Wiley & Sons, Ltd., Chichester, UK, 1989, pp. 511–554, <http://dx.doi.org/10.1002/0470025107.ch8> (accessed January 08.01.14).
 - [48] K. Pielichowski, K. Flejtuch, Non-oxidative thermal degradation of poly(ethylene oxide): kinetic and thermoanalytical study, *J. Anal. Appl. Pyrolysis* 73 (2005) 131–138, <http://dx.doi.org/10.1016/j.jaap.2005.01.003>
 - [49] N.S. Choong Kwet Yive, R.J.P. Corriu, D. Leclercq, P.H. Mutin, A. Vioux, Silicon carbonitride from polymeric precursors: thermal cross-linking and pyrolysis of oligosilazane model compounds, *Chem. Mater.* 4 (1992) 141–146, <http://dx.doi.org/10.1021/cm00019a029>
 - [50] B. Gardelle, S. Duquesne, C. Vu, S. Bourbigot, Thermal degradation and fire performance of polysilazane-based coatings, *Thermochim. Acta* 519 (2011) 28–37, <http://dx.doi.org/10.1016/j.tca.2011.02.025>
 - [51] D. Bahloul, M. Pereira, P. Goursat, N.S.C.K. Yive, R.J.P. Corriu, Preparation of silicon carbonitrides from an organosilicon polymer: I. Thermal decomposition of the cross-linked polysilazane, *J. Am. Ceram. Soc.* 76 (1993) 1156–1162, <http://dx.doi.org/10.1111/j.1151-2916.1993.tb03734.x>
 - [52] W.R. Gombotz, W. Guanghui, T.A. Horbett, A.S. Hoffman, Protein adsorption to poly(ethylene oxide) surfaces, *J. Biomed. Mater. Res.* 25 (1991) 1547–1562, <http://dx.doi.org/10.1002/jbm.b.820251211>
 - [53] S.R. Benhabbour, H. Sheardown, A. Adronov, Protein resistance of PEG-functionalized dendronized surfaces: effect of PEG molecular weight and dendron generation, *Macromolecules* 41 (2008) 4817–4823, <http://dx.doi.org/10.1021/ma8004586>
 - [54] B. Zhu, T. Eurell, R. Gunawan, D. Leckband, Chain-length dependence of the protein and cell resistance of oligo(ethylene glycol)-terminated self-assembled monolayers on gold, *J. Biomed. Mater. Res.* 56 (2001) 406–416, [doi:10.1002/1097-4636\(20010905\)56:3<406::AID-JBM1110>3.0.CO;2-R](http://dx.doi.org/10.1002/1097-4636(20010905)56:3<406::AID-JBM1110>3.0.CO;2-R)
 - [55] S.I. Jeon, J.H. Lee, J.D. Andrade, P.G. De Gennes, Protein-surface interactions in the presence of polyethylene oxide: I. Simplified theory, *J. Colloid Interface Sci.* 142 (1991) 149–158, [http://dx.doi.org/10.1016/0021-9797\(91\)90043-8](http://dx.doi.org/10.1016/0021-9797(91)90043-8)